



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2011
Trial Higher School
Certificate

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 2 Hours
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- All necessary working should be shown in every question, if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- All answers to be given in simplified exact form unless otherwise stated.

Total Marks – 84

- Attempt all questions 1-7
- All questions are of equal value.
- Start each new question in a separate answer booklet.
- Hand in your answers in 7 separate bundles: Questions 1, 2, 3, 4, 5, 6 and 7.
- The mark value of each question is given in the right margin.

Examiner: *R Boros*

Start a new booklet.

Question 1 (12 marks).

Marks.

a) Solve $x(3-2x) > 0$. 2

b) Find $\frac{d}{dx}(e^{-x} \cos^{-1} x)$. 2

c) The remainder when $x^3 + ax^2 - 3x + 5$ is divided by $(x + 2)$ is 11. Find the value of a . 2

d) Using the table of standard integrals, find the exact value of: 2

$$\int_0^{\frac{\pi}{8}} \sec 2x \tan 2x \, dx.$$

e) Solve for x , $\frac{x^2 - 9}{x} \geq 0$. 2

f) Find $\int_0^2 \frac{1}{4+x^2} dx$, leaving your answer in exact form. 2

End of Question 1.

Start a new booklet.

Question 2 (12 marks).	Marks.
a) Use the substitution $x = \ln u$ to find: $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx.$	3
b) Use one application of Newton's method to find an approximation to the root of the equation $\cos x = x$ near $x = \frac{1}{2}$. Give your answer correct to 2 decimal places.	2
c) The curves $y = e^{2x}$ and $y = 1 + 4x - x^2$ intersect at the point $(0,1)$. Find the angle, to the nearest minute, between the 2 curves at this point of intersection.	3
d) Prove that $\frac{2}{\tan A + \cot A} = \sin 2A$.	2
e) Find the derivative of $\cos^3 x^\circ$.	2

End of Question 2.

Start a new booklet.

Question 3 (12 marks).

Marks.

- a)**
- (i) Expand $\cos(\alpha + \beta)$ 1
 - (ii) Show that $\cos 2\alpha = 1 - 2\sin^2 \alpha$ 1
 - (iii) Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$ 1
- b)** If $\alpha = \tan^{-1}\left(\frac{5}{12}\right)$ and $\beta = \cos^{-1}\left(\frac{4}{5}\right)$, calculate the exact value of $\tan(\alpha - \beta)$. 2
- c)** $A(-1, 7)$ and $B(5, -2)$ are 2 points. Point P divides AB in the ratio $k : 1$.
- (i) Write down the coordinates of P in terms of k . 2
 - (ii) If P lies on the line $5x - 4y - 1 = 0$, find the ratio of $AP:PB$. 1
- d)** Use mathematical induction to prove that: 3
- $$1 \times 1! + 2 \times 2! + 3 \times 3! + \dots + n \times n! = (n+1)! - 1$$
- where n is a positive integer.
- e)** State the domain of $y = 2 \sin^{-1}(1 - x)$. 1

End of Question 3.

Start a new booklet.

Question 4 (12 marks).

Marks.

a) $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ are 2 points on the parabola $x^2 = 4ay$.

(i) Show that the equation of PQ is given by: $y - \frac{1}{2}(p+q)x + apq = 0$ 2

(ii) Find the condition that PQ passes through the point $(0, -a)$. 1

(iii) If the focus of the parabola is S and PQ passes through $(0, -a)$, 2

prove that $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$.

b) A family consists of a father, mother, 3 girls and 4 boys.

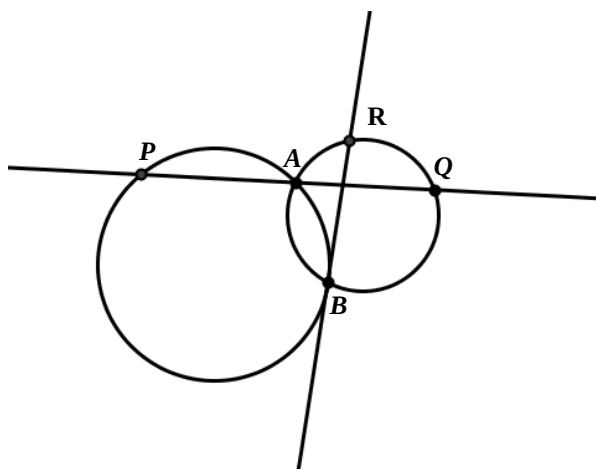
(i) If the family is seated at random along a bench, find the probability 1
that the parents are at the ends, and the boys and girls are seated
alternately between them.

(ii) If the same family is seated randomly at a round table, find the 1
probability that the parents are separated by exactly two seats, and
these seats are both occupied by boys.

Question 4 continued on next page.

Question 4 continued.**Marks.**

- c) Two circles cut at A and B . A line through A meets one circle at P . Also, BR is a tangent to the circle ABP and R lies on the circle ABQ .



- (i) Copy the diagram showing the above information.
- (ii) Prove that $PB \parallel QR$. 2
- d) One root of $x^3 + px^2 + qx + r = 0$ equals the sum of the two other roots. 3
- Prove that $p^3 + 8r = 4pq$.

End of Question 4.

Start a new booklet.

Question 5 (12 marks).

Marks.

- a)** The area bounded by the curve $y = \sin^{-1} x$, the y -axis and $y = \frac{\pi}{2}$ is rotated about the y -axis.

- (i) Show that the volume of the solid so formed is given by:

1

$$\pi \int_0^{\frac{\pi}{2}} \sin^2 y \, dy.$$

- (ii) Hence, find the exact volume of this solid.

2

- b)** The area of an equilateral triangle is increasing at the rate of $4 \text{ cm}^2/\text{s}$.

- (i) If x is the length of the side of the triangle, find an expression for the area of the triangle.

1

- (ii) Find the exact rate of increase of the side of the triangle, when it has a side length of 2 cm.

2

Question 5 continued on next page.

Question 5 continued.**Marks.**

- c)** (i) Find the largest possible domain of positive values for which
 $f(x) = x^2 - 6x + 13$ has an inverse. 1
- (ii) Find the equation of the inverse function $f^{-1}(x)$. 2
- d)** (i) Prove that $\frac{d}{dx}\left(\frac{1}{2}v^2\right) = \ddot{x}$. 1
- (ii) A cat moving in a straight line has an acceleration given by
 $\ddot{x} = x(8 - 3x)$, where x is the displacement in metres from a fixed point
 O . Initially the cat is at the origin, O , and has a speed of 4 m/s. Find
the cat's speed when it is 1 m on the positive side of O . 2

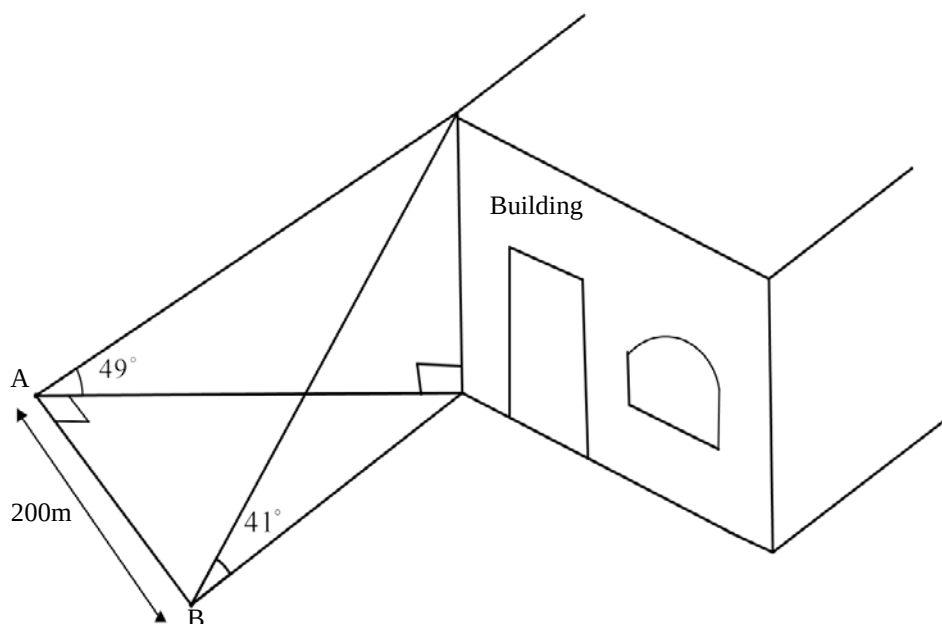
End of Question 5.

Start a new booklet.

Question 6 (12 marks).

Marks.

- a) Given $f(x) = 3\cos^{-1}(\sin 2x) - 2\sin^{-1}(\cos 3x)$, show that $f(x)$ is a constant function by finding $f'(x)$. 2
- b) From point A, due south of a building, the angle of elevation to the top of the building is 49° . Two hundred metres due east of A, at point B, the angle of elevation is found to be 41° as shown in the diagram below. Determine, to the nearest metre, the height of the building. 3



Question 6 continued on next page.

Question 6 continued.**Marks.**

- c) A particle is oscillating in simple harmonic motion such that its displacement x metres from a given origin O satisfies the equation

$$\frac{d^2x}{dt^2} = -4x, \text{ where } t \text{ is the time in seconds.}$$

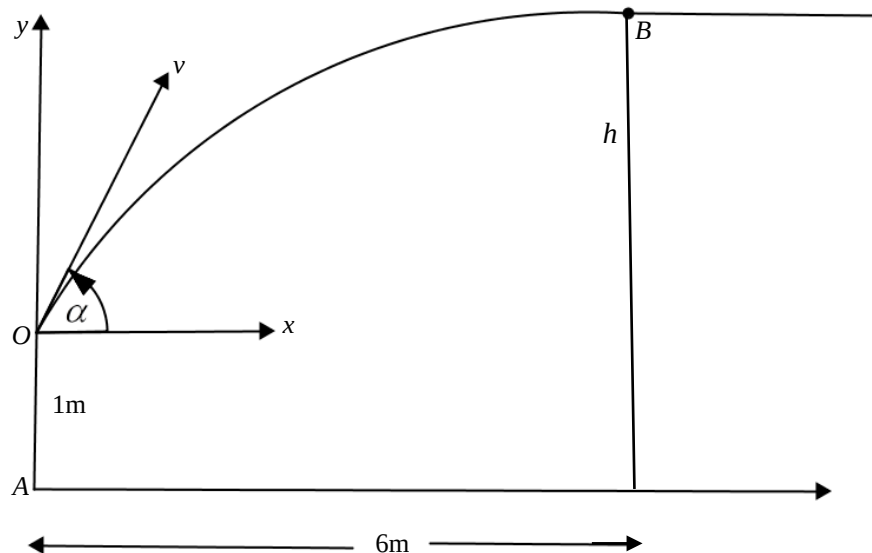
- (i) Show that $x = a \cos(2t + \beta)$ is a possible equation of motion for the particle, where a and β are constants. 1
- (ii) The particle is observed at time $t = 0$ to have a velocity of 2 m/s and a displacement from the origin of 4 m. Find the amplitude of the oscillation. 2
- (iii) Determine the maximum velocity of the particle. 1
- d) Using $t = \tan \frac{\theta}{2}$, find the general solution in radians to $\sin \theta - \cos \theta = 1$. 3

End of Question 6.

Start a new booklet.

Question 7 (12 marks).

Marks.



- a) A girl, 1 metre tall, throws a frisbee from a point O , with velocity v m/s at an angle α with the horizontal. It strikes the wall of a building at the highest point B of its trajectory. It takes $1\frac{1}{2}$ seconds to travel from O to B and the wall is 6 metres away from the girl. Taking the coordinate axes from O as shown and $g \approx 10 \text{ m/s}^2$.

- (i) Prove that at any time t , the position of the frisbee is given by: 2

$$x = vt \cos \alpha \text{ and } y = vt \sin \alpha - 5t^2.$$

- (ii) Show that $v \cos \alpha = 4$ and $v \sin \alpha = 15$. 2

- (iii) Determine the initial velocity v , in exact form, and the angle of projection α to the nearest degree. 2

- (iv) Find the height h of the building, correct to 2 decimal places. 1

Question 7 continued on next page.

Question 7 continued.**Marks.**

- b)** $P(x, y)$ is a point on the curve $y = e^{-x^2}$, where $x > 0$, O is the origin, and the perpendiculars from P to the x -axis and y -axis meet at A and B respectively.

(i) Show that the maximum area of the rectangle $OAPB$ is $\frac{1}{\sqrt{2e}}$. 2

(ii) Show that the minimum length of OP is $\sqrt{\frac{1 + \ln 2}{2}}$. 3

End of Question 7.**End of Examination.**

Standard Integrals.

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

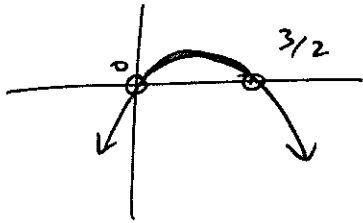
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$

SBMS TMSC EXT 1 2011

Question 1

(a) $x(3-2x) > 0$



$$0 < x < \frac{3}{2}$$

(b) $\frac{d}{dx}(e^{-x} \cos^{-1} x) = e^{-x} \cdot \frac{-1}{\sqrt{1-x^2}} + \cos^{-1} x \cdot -e^{-x}$
 $= -e^{-x} \left(\cos^{-1} x + \frac{1}{\sqrt{1-x^2}} \right)$

(c) $P(x) = x^3 + ax^2 - 3x + 5$

$$P(-2) = (-2)^3 + a(-2)^2 - 3(-2) + 5 = 11$$

$$-8 + 4a + 6 + 5 = 11$$

$$4a = 8$$

$$a = 2$$

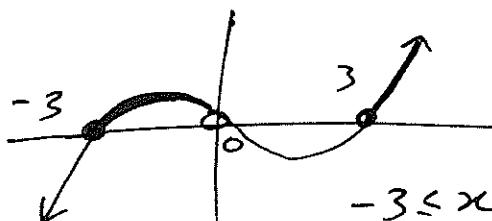
(d) $\int_0^{\frac{\pi}{8}} \sec 2x + \tan 2x \, dx = \left[\frac{1}{2} \sec 2x \right]_0^{\frac{\pi}{8}}$
 $= \frac{1}{2} \sec 2\left(\frac{\pi}{8}\right) - \frac{1}{2} \sec 2(0)$
 $= \frac{1}{2} \sec \frac{\pi}{4} - \frac{1}{2} \sec 0$
 $= \frac{1}{2}(\sqrt{2}) - \frac{1}{2}(1)$
 $= \frac{1}{2}(\sqrt{2} - 1)$

(e) $\frac{x^2 - 9}{x} \geq 0$

$$x \neq 0$$

$$x(x^2 - 9) \geq 0$$

$$x(x-3)(x+3) \geq 0$$



$$-3 \leq x < 0, x \geq 3$$

$$\begin{aligned} (f) \quad & \int_0^2 \frac{1}{4+x^2} dx \\ &= \left[\frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) \right]_0^2 \\ &= \frac{1}{2} \tan^{-1}\left(\frac{2}{2}\right) - \frac{1}{2} \tan^{-1}\left(\frac{0}{2}\right) \\ &= \frac{1}{2} \left(\frac{\pi}{4}\right) - \frac{1}{2}(0) \\ &= \frac{\pi}{8}. \end{aligned}$$

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a)

$$\begin{aligned} & \int \frac{e^x}{\sqrt{1-e^{2x}}} \cdot dx \\ &= \int \frac{1}{\sqrt{1-u^2}} \cdot du \\ &= \sin^{-1} u + C \\ &= \sin^{-1}(e^x) + C \end{aligned}$$

$$\begin{aligned} x &= \ln u \\ e^x &= u \\ du &= e^x \cdot dx \end{aligned}$$

b) $f(x) = x - \cos x$
 $f'(x) = 1 + \sin x$

$$\begin{aligned} x_2 &= \frac{1}{2} - \frac{f\left(\frac{1}{2}\right)}{f'\left(\frac{1}{2}\right)} \\ x_2 &= 0.5 + \frac{-0.37758}{1.479426} \\ x_2 &= 0.76 \end{aligned}$$

c) $y = e^{2x}$
 $y' = 2e^{2x}$
 $y = 1 + 4x - x^2$
 $y' = 4 - 2x$
 At $(0, 1)$
 $m_1 = 2, \quad m_2 = 4$
 $\tan \theta = \left| \frac{2 - 4}{1 + 2(4)} \right|$
 $= \left| -\frac{2}{9} \right|$
 $\therefore \theta = 12^\circ 32'$

d)

$$\begin{aligned} \frac{2}{\tan A + \cot A} &= \sin 2A \\ LHS &= \frac{2}{\frac{\sin A}{\cos A} + \frac{\cos A}{\sin A}} \\ &= \frac{2}{\frac{\sin^2 A + \cos^2 A}{\sin A \cos A}} \\ &= \frac{2}{1} \\ &= 2 \sin A \cos A \\ &= \sin 2A \\ &= RHS \end{aligned}$$

e)

$$\begin{aligned} & \frac{d}{dx} [\cos^3 x^\circ] \\ &= \frac{d}{dx} \left[\cos^3 \left(\frac{\pi x}{180} \right) \right] \\ &= 3 \cos^2 \left(\frac{\pi x}{180} \right) \cdot -\frac{\pi}{180} \sin \left(\frac{\pi x}{180} \right) \\ &= -\frac{\pi}{60} \cos^2 \left(\frac{\pi x}{180} \right) \sin \left(\frac{\pi x}{180} \right) \end{aligned}$$

QUESTION 3 $\times 1$

$$\begin{aligned}\cos(\alpha + \beta) &= \cos\alpha \cos\beta - \sin\alpha \sin\beta \\ &= \cos(\alpha + \alpha) = \cos^2\alpha - \sin^2\alpha \\ &= 1 - \sin^2\alpha - \sin^2\alpha \\ &= 1 - 2\sin^2\alpha\end{aligned}$$

$$\begin{aligned}(ii) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} \\ &= \lim_{x \rightarrow 0} \frac{2\sin^2 x}{x^2} \\ &= \lim_{x \rightarrow 0} 2 \frac{\sin x}{x} \cdot \frac{\sin x}{x} \\ &= 2\end{aligned}$$

$$\begin{aligned}(b.2) \tan(\alpha - \beta) &= \frac{\frac{5}{12} - \frac{3}{4}}{1 + \frac{5}{12} \cdot \frac{3}{4}} \\ &= \frac{-16}{3} \text{ or } -0.254\end{aligned}$$

$$(c) \quad P_x = \frac{5k-1}{k+1} \quad P_y = \frac{-2k+7}{k+1}$$

(i)

$$(ii) \quad \frac{25k-5}{k+1} = \frac{-8k+28}{k+1} - 1 = 0$$

$$33k - 33 = k + 1$$

$$32k = 34$$

$$k = \frac{34}{32} = \frac{17}{16}$$

$$k = \frac{17}{16}; 1 = 17:16$$

(d) if $n=1$

$$LHS = 1 \times 1! \quad RHS = 2 - 1$$

$S(1)$ is true.

Assume $S(k)$ true

$$1 \times 1! + 2 \times 2! + \dots + k \times k! = (k+1)! - 1$$

$S(k+1)$ is

$$1 \times 1! + 2 \times 2! + k \times k! + (k+1)(k+1)! = (k+2)! - 1$$

$$\begin{aligned}LHS &= (k+1)! - 1 + (k+1)(k+1)! \text{ by assumption} \\ &= (k+1)! [1 + k+1] - 1 \\ &= (k+2)(k+1)! - 1 \\ &= (k+2)! - 1 = RHS\end{aligned}$$

$\therefore S(k+1)$ is true if $S(k)$ is true

$\therefore$ by Mathematical Induction, $S(n)$ is true for any integer $n \geq 1$

$$(e) \quad -1 \leq x \leq 1$$

$$1 \geq -x \geq -1$$

$$2 \geq 1-x \geq 0$$

$$0 \leq 1-x \leq 2$$

$$\sqrt{\quad} = \frac{1}{2}m$$

4(a)

$$(i) \quad m_{PQ} = \frac{ap^2 - aq^2}{2ap - 2aq} = \frac{\cancel{a}(p^2 - q^2)}{2\cancel{a}(p - q)}$$

$$= \frac{(p - q)(p + q)}{2(p - q)}$$

$$\Rightarrow m_{PQ} = \frac{1}{2}(p + q) \quad \checkmark$$

Then Equation of PQ: $y - y_1 = m(x - x_1)$

$$y - ap^2 = \frac{1}{2}(p + q)(x - 2ap)$$

$$y - ap^2 = \frac{1}{2}(p + q)x - ap(p + q)$$

$$y - ap^2 = \frac{1}{2}(p + q)x - ap^2 - apq$$

$$\Rightarrow y = \frac{1}{2}(p + q)x - apq$$

$$\therefore y - \frac{1}{2}(p + q)x + apq = 0 \quad \checkmark$$

(2)

(ii) PQ passes through $(0, -a)$

$$\Rightarrow -a - \frac{1}{2}(p + q)0 + apq = 0 \quad \checkmark$$

$$-a + apq = 0$$

$$a(pq - 1) = 0$$

$$\Rightarrow a = 0 \text{ or } pq = 1$$

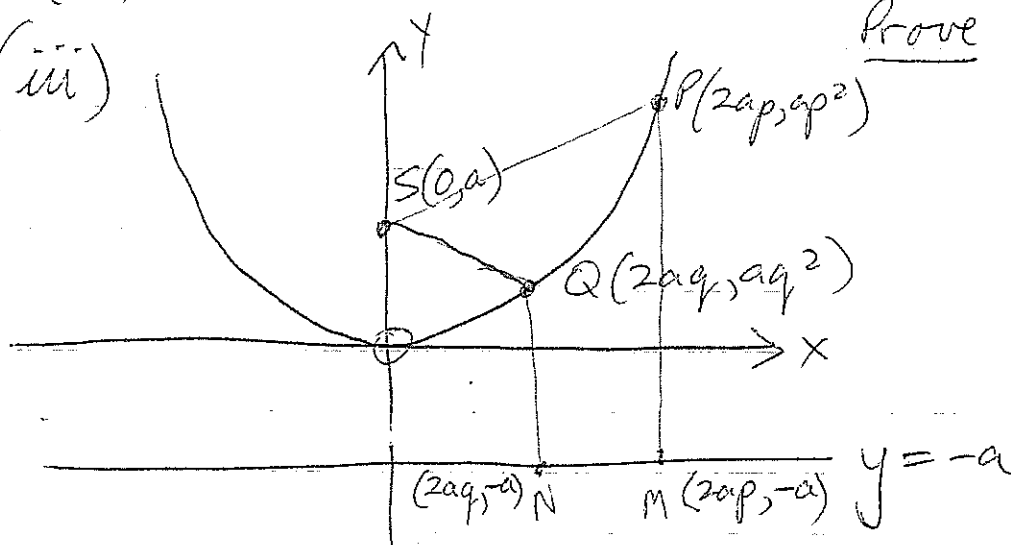
Not a parabola

$$\therefore \underline{pq = 1} \quad \checkmark$$

(1)

4 (a)

(iii)



Prove $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$.

From defn of a parabola,

$$SP = PM \quad \text{and} \quad SQ = QN$$

$$\text{Now } PM^2 = (ap^2 + a)^2 + (2ap - 2ap)^2$$

$$\Rightarrow PM = ap^2 + a = a(p^2 + 1) = SP$$

$$\text{Also, } QN = aq^2 + a = a(q^2 + 1) = SQ$$

$$\text{Then } \frac{1}{SP} + \frac{1}{SQ}$$

$$= \frac{1}{a(p^2 + 1)} + \frac{1}{a(q^2 + 1)}$$

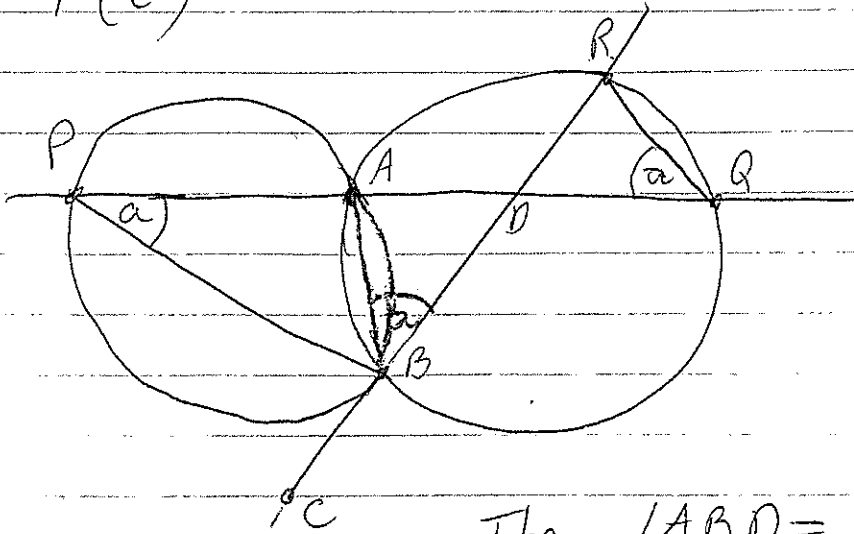
$$= \frac{q^2 + 1 + p^2 + 1}{a(p^2 + 1)(q^2 + 1)} = \frac{p^2 + q^2 + 2}{a(p^2 q^2 + p^2 + q^2 + 1)}$$

But $pq = 1$ (from ii)

$$\Rightarrow \frac{1}{SP} + \frac{1}{SQ} = \frac{p^2 + q^2 + 2}{a(p^2 + q^2 + 2)} = \frac{1}{a} \quad \#$$

$$\therefore P(\text{M and F separated by 2 boys in circle}) = \frac{2880}{40320} = \frac{1}{14} \checkmark$$

4(c)



Prove $PB \parallel QR$

Construct AB, BP, RQ

Then $\angle ABD = \angle APB$ (Alternate $\angle$ Theorem) ✓
 let these angles $= a$.

Also $\angle RQA = \angle RBA$ (Angles at circumference standing on same arc are equal) ✓

but $\angle RBA = \angle ABD = a$ (same angle) ✓

$\therefore \angle RQA = \angle ABD = a$

$\therefore PB \parallel QR$ (Alternate angles are equal) ✓

2

$$4(d) \quad x^3 + px^2 + qx + r = 0$$

given α, β, r Let roots be $\alpha, \beta, \alpha + \beta$

$$\text{Then } 2(\alpha + \beta) = -p \quad (\text{sum of roots} = -\frac{b}{a})$$

$$\alpha + \beta = -\frac{p}{2}$$

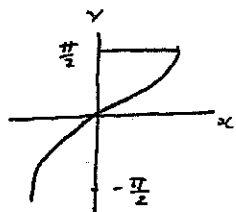
But $(\alpha + \beta)$ is a root.

$$\Rightarrow p\left(-\frac{p}{2}\right) = \frac{-p^3}{8} + \frac{p^3}{4} - \frac{pq}{2} + r = 0$$

$$\times 8 \Rightarrow -p^3 + 2p^3 - 4pq + 8r = 0.$$

$$\Rightarrow 4pq = p^3 + 8r \quad \#$$

(a) $y = \sin^{-1} x$
 $\therefore x = \sin y$



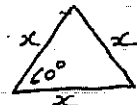
(i)
$$V = \pi \int_0^{\pi/2} x^2 \cdot dy$$

$$= \pi \int_0^{\pi/2} \sin^2 y \cdot dy$$

$$= \pi \int_0^{\pi/2} \frac{1}{2} (1 - \cos 2y) dy$$

$$= \frac{\pi}{2} \left[y - \frac{\sin 2y}{2} \right]_0^{\pi/2}$$

(ii) $= \frac{\pi^2}{4}$

(b)  (i) $A = \frac{1}{2} \cdot x^2 \cdot \sin 60$
 $= \frac{\sqrt{3} x^2}{4}$

(ii) $A = \frac{\sqrt{3} x^2}{4}$
 $\frac{dA}{dt} = \frac{\sqrt{3} x}{2}$

$\frac{dA}{dt} = \frac{dA}{dx} \cdot \frac{dx}{dt}$

$4 = \frac{\sqrt{3} x}{2} \cdot \frac{dx}{dt}$

at $x=2$; $\frac{dx}{dt} = \frac{4}{\sqrt{3}} = \frac{4\sqrt{3}}{3}$

(2) $f(x) = x^2 - 6x + 13$

$f(x) = (x-3)^2 + 4$

Vertex at (3,4)

Largest domain of positive values $\{x: x \geq 3\}$

For inverse function $x = (y-3)^2 + 4$

$(y-3)^2 = x-4$

$y-3 = \sqrt{x-4}$

$y = 3 + \sqrt{x-4}$

$\therefore f^{-1}(x) = 3 + \sqrt{x-4}$

(2)(i) Show $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \ddot{x}$

$\ddot{x} = \frac{dv}{dt}$

or $\frac{d(\frac{1}{2} v^2)}{dx} = \frac{d(\frac{1}{2} v^2)}{dv} \cdot \frac{dv}{dt} \cdot \frac{dt}{dx}$

$= \frac{dv}{dx} \cdot \frac{dx}{dt}$

$= v \cdot \frac{dv}{dt} = \frac{1}{2} \frac{d(v^2)}{dt}$

$= \frac{dv}{dt} \cdot v$

$= \frac{d(v^2)}{dt}$

$= \frac{d(\frac{1}{2} v^2)}{dx} \cdot \frac{dx}{dt}$

$= \ddot{x}$

$= \frac{d(\frac{1}{2} v^2)}{dx}$

(ii) $\ddot{x} = 8x - 3x^2$

$\therefore \frac{d(\frac{1}{2} v^2)}{dx} = 8x - 3x^2$

By integration $\frac{1}{2} v^2 = 4x^2 - x^3 + C$

at $t=0, x=0, v=4$; $\therefore 8 = C$

$\therefore \frac{1}{2} v^2 = 4x^2 - x^3 + 8$

$v^2 = 8x^2 - 2x^3 + 16$

at $x=1$;

$v^2 = 8 - 2 + 16$

$v^2 = 22$

$v = \pm \sqrt{22}$

Speed (has no direction) $Ans = \sqrt{22}$

QUESTION 6 X1

(a) $y = 3 \cos^{-1}(\sin 2x) - 2 \sin^{-1}(\cos 3x)$

$$y' = -3 \frac{x 2 \cos 2x}{\sqrt{1 - \sin^2 2x}} - \frac{2x - 3 \sin 3x}{\sqrt{1 - \cos^2 3x}}$$

$$= -\frac{6 \cos 2x}{\cos 2x} + \frac{6 \sin 3x}{\sin 3x}$$

$$= 0$$

$\therefore y$ constant

(b) $\tan 49 = \frac{h}{a} \quad \tan 41 = \frac{h}{b}$

$$200^2 + \frac{h^2}{\tan^2 49} = \frac{h^2}{\tan^2 41}$$

$$h^2 \left(\frac{1}{\tan^2 41} - \frac{1}{\tan^2 49} \right) = 200^2$$

$$h^2 = 200^2 \div \left(\frac{1}{\tan^2 41} - \frac{1}{\tan^2 49} \right)$$

$$h \approx 265$$

(c) (i) $x = a \cos(2t + \beta)$

$$\dot{x} = -2a \sin(2t + \beta)$$

$$\ddot{x} = -4a \cos(2t + \beta)$$

$$= -4x$$

Q(ii) $t = 0 \quad v = 0 \quad x = 4$

$$v^2 = n^2 (a^2 - x^2)$$

$$4 = 4 (a^2 - 16)$$

$$a = \sqrt{17}$$

(iii) max velocity = an
 $= 2\sqrt{17}$

(iv) $t = \tan \frac{\theta}{2}$

$$\sin \theta = \frac{2t}{1+t^2} \quad \cos \theta = \frac{1-t^2}{1+t^2}$$

$$\sin \theta - \cos \theta = \frac{2t - 1 + t^2}{1+t^2} = 1$$

$$2t - 1 + t^2 = 1 + t^2$$

$$2t = 2$$

$$t = 1$$

$$\tan \frac{\theta}{2} = 1$$

$$\theta = \pi$$

or

$$\theta = (2n+1)\pi + \frac{\pi}{2} \quad \therefore$$

$n \in \mathbb{Z}$

(X1)

Q7.

(1)

$$\ddot{x} = 0$$

$$\dot{x} = c_1$$

$$\therefore \dot{x} = v \cos \alpha$$

when $t=0$

$$\therefore \dot{x} = v \cos \alpha$$

$$x = vt \cos \alpha + c_3$$

$$x=0, t=0$$

$$\therefore c_3 = 0$$

$$x = vt \cos \alpha$$

$$\ddot{y} = -10$$

$$\dot{y} = -10t + c_2$$

$$\dot{y} = v \sin \alpha$$

when $t=0$

$$\therefore c_2 = v \sin \alpha$$

hence.

$$\dot{y} = -10t + v \sin \alpha$$

$$y = -5t^2 + vt \sin \alpha + c_4$$

$$\text{when } t=0, y=0$$

$$\therefore c_4 = 0$$

hence.

$$y = vt \sin \alpha - 5t^2$$

(ii) when $t = 1.5$ $x = 6$ & $\dot{y} = 0$

$$\therefore 6 = v \times \frac{3}{2} \cos \alpha$$

$$v \cos \alpha = 4$$

$$0 = -10 \times \frac{3}{2} + v \sin \alpha$$

$$v \sin \alpha = 15$$

(iii) $\sin^2 \alpha + \cos^2 \alpha = 1$

$$\therefore \left(\frac{15}{v}\right)^2 + \left(\frac{4}{v}\right)^2 = 1$$

$$225 + 16 = v^2$$

$$v = \sqrt{241}$$

m/s.

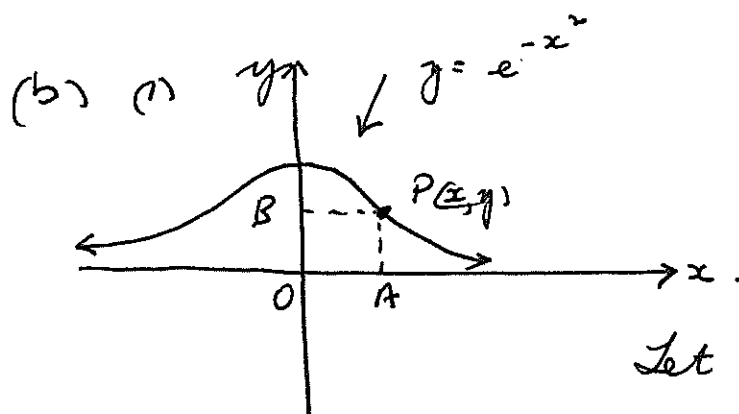
$$\cos \alpha = \frac{4}{\sqrt{241}}$$

$$\therefore \alpha = 75^\circ$$

$$h = 2t \sin \alpha - 5t^2 + 1$$

$$= 2 \times 1.5 \times \frac{15}{\sqrt{241}} - 5 \times \frac{9}{4} + 1$$

$$= 12.25$$



Let $A = \text{Area of } OAPB$

$$= xy.$$

$$= x e^{-x^2}$$

$$\frac{dA}{dx} = x \cdot -2x e^{-x^2} + e^{-x^2}$$

$$= e^{-x^2} (1 - 2x^2)$$

$$\text{Set } \frac{dA}{dx} = 0$$

$$1 - 2x^2 = 0$$

$$x^2 = \frac{1}{2}$$

$$x = \frac{1}{\sqrt{2}} \quad (\text{NB } x > 0)$$

$$\therefore A = \frac{1}{\sqrt{2}} e^{-\left(\frac{1}{\sqrt{2}}\right)^2}$$

$$= \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{e}}$$

$$= \frac{1}{\sqrt{2e}}$$

* [MUST TEST
FOR MAX.
USING NUMBERS]

(ii) $l_{OP} = \sqrt{x^2 + y^2}$

$$= \sqrt{x^2 + (e^{-x^2})^2}$$

$$= (x^2 + e^{-2x^2})^{\frac{1}{2}}$$

$$l' = \frac{1}{2} (x^2 + e^{-2x^2})^{-\frac{1}{2}} \times (2x - 4x e^{-2x^2})$$

$$= \frac{2x(1 - 2e^{-2x^2})}{2\sqrt{x^2 + e^{-2x^2}}}$$

let $l' = 0$

ie. $2x(1 - 2e^{-2x^2}) = 0$

$x = 0$ OR $e^{-2x^2} = \frac{1}{2}$

↓

then $OP = 1$.

$$e^{-2x^2} = \frac{1}{2}$$

$$e^{2x^2} = 2$$

$$2x^2 = \ln 2$$

$$x^2 = \frac{\ln 2}{2}$$

$$x = \sqrt{\frac{\ln 2}{2}}, (x > 0)$$

$$\therefore l = \sqrt{\frac{\ln 2}{2} + e^{-2 \frac{\ln 2}{2}}}$$

$$= \sqrt{\frac{\ln 2}{2} + e^{-\ln 2}}$$

$$= \sqrt{\frac{\ln 2}{2} + e^{\ln \frac{1}{2}}}$$

$$= \sqrt{\frac{\ln 2}{2} + \frac{1}{2}}$$

$$= \sqrt{\frac{1 + \ln 2}{2}}$$

(NB
MUST TEST
FOR MIN
USING
VALUES)